



Mohammed Raihan Soniwala

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EDUCATION AND TRAINING

[07/2019 – 04/2023]

Bachelor in Electronics and Communication Engineering

Gujarat Technological University

City: Ahmedabad | **Country**: India | **Final grade**: 1.78

Relevant coursework : *Embedded Systems, Internet of Things, Micro-Processor and Micro-Controller Technology, Power Electronics, Problem Solving Programming, Probability and Statistics, Signal and Systems.*

[04/2023 – Current]

Masters in Electrical Engineering

University of Stuttgart

City: Stuttgart | **Country**: Germany |

WORK EXPERIENCE

[08/2022 – 01/2024]

Embedded Engineer

Seagull GNSS PVT LTD

City: Ahmedabad | **Country**: India

- Developed and optimized Embedded C applications for microcontrollers, delivering reliable real-time performance while ensuring efficient memory utilization.
- Designed and optimized PCB layouts using KiCad and Altium Designer, ensuring robust signal integrity and minimizing electromagnetic interference for enhanced circuit performance.

PROJECTS

Voice Chat GPT

Created the "VoiceChat GPT" project involved leveraging the combined power of Bluetooth communication, Python programming, and Raspberry Pi hardware to build a sophisticated and conservative conversational AI device with voice interaction and speaker output.

Link: <https://github.com/rayy2002/VoiceChat-GPT>

WiFi Control Android Application

Developed an application in Flutter that allows users to control an ESP's WiFi Station by connecting to its Access Point.

Link: <https://github.com/rayy2002/FlutterESP-Wi-Fi-Control>

RTOS Functionality on ESP32S3

Implemented RTOS functionalities using Vanilla FreeRTOS on various microcontrollers at Seagull GNSS Pvt Ltd, facilitating communication using I2C, CAN, UART, WiFi, and RF protocols to enhance real-time data processing and communication efficiency.

Machine Learning Projects

Link: <https://www.kaggle.com/raihansoniwala/code>

EXTRACURRICULAR ACTIVITIES

Electrical Engineer | KSat e. V. (small satellite student society at the University of Stuttgart)

- Collaborated with the EPS (Electrical Power Supply) team to manage all power supply and battery related hardware tasks for satellite development, ensuring reliable and efficient power management systems.
- Engineered an interface between VORAGO microcontrollers and MRAM (Magnetoresistive RAM) to securely store up-time data, enabling seamless clock recovery post-battery blackout, thereby enhancing system resilience and ensuring data integrity.

Link: <https://www.ksat-stuttgart.de/en/>

IoT Engineer | Team Falcon (Robotics society at L.J .I.E.T, Ahmedabad)

- Collaborated with a team to design and build a competition-ready robot for the ABU Robocon, utilizing ESP controllers for precise motor control and seamless coordination of movements, while successfully implementing electrical installations to meet task-specific requirements.

DIGITAL SKILLS

My Digital Skills

Embedded Systems

Embedded C | C++ | C | PCB Design | UART | I2C | RTOS | SPI | Linux | STM32 | Eclipse | Device Testing and Debugging | ARM | Firmware | MATLAB | Good Communication Skills

Artificial Intelligence

Python | Algorithms | Computer Vision | Data Analysis | Problem Solving | Data visualization | Natural language processing | Keras | Tensorflow | Data Modelling | Data Structures

Management skills

Teamwork | Patience | Conflict resolution | Collaboration | Problem solving | Decision-making | Adaptability | interpersonal skills

CERTIFICATIONS

Bare metal C programming on STM32

[14/11/2024]

Advanced Learning Algorithms by Andrew NG on Coursera

[14/10/2024]

Supervised Machine Learning: Regression and Classification by Andrew NG on Coursera

HOBBIES AND INTERESTS

Interests

Football | LEGO | Exploring new microcontrollers | Debugging